29/07/2026

1. Description

1.1. Project

Project Name	nexus_first
Board Name	NUCLEO-H7S3L8
Generated with:	STM32CubeMX 6.18.0
Date	07/29/2026

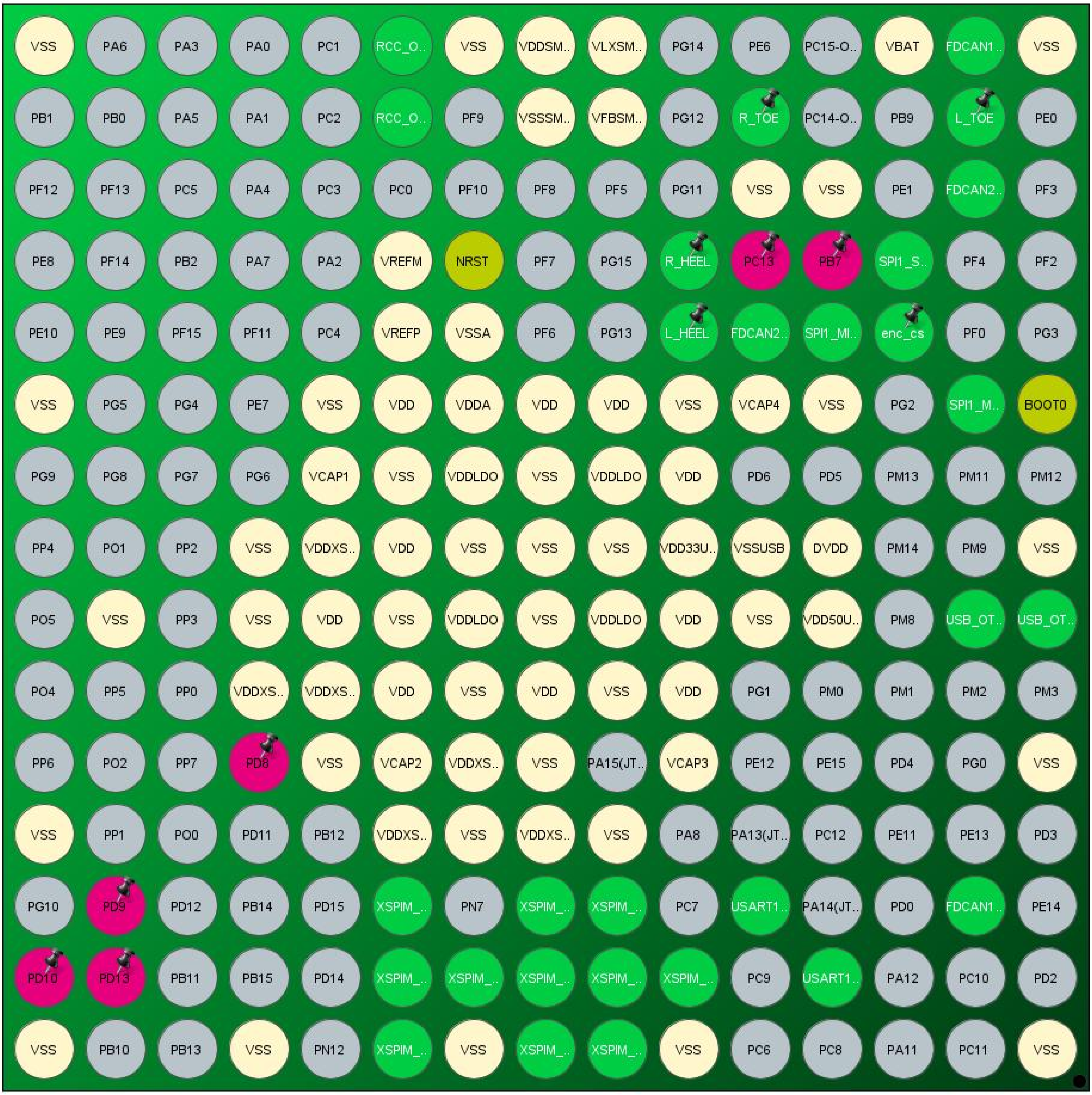
1.2. MCU

MCU Series	STM32H7RS
MCU Line	STM32H7R3/7S3
MCU name	STM32H7S3L8Hx
MCU Package	TFBGA225 OCTO SMPS
MCU Pin number	225

1.3. Core(s) information

Core(s)	ARM Cortex-M7

2. Pinout Configuration



TFBGA225 OCTO SMPS (Bottom view - Rotated +90°)

3. Pins Configuration

Pin Number TFBGA225 OCTO SMPS	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
A1	VSS	Power		
A6	BOOT0	Boot		
A8	VSS	Power		
A9	PM6	I/O	USB_OTG_HS_DP	
A11	VSS	Power		
A15	VSS	Power		
B1	PB8	I/O	FDCAN1_RX	
B2	PE2 *	I/O	GPIO_Input	L_TOE
B3	PB5	I/O	FDCAN2_RX	
B6	PD7	I/O	SPI1_MOSI	
B9	PM5	I/O	USB_OTG_HS_DM	
B13	PD1	I/O	FDCAN1_TX	
C1	VBAT	Power		
C4	PB3(JTDO/TRACESWO)	I/O	SPI1_SCK	
C5	PF1 *	I/O	GPIO_Output	enc_cs
D3	VSS	Power		
D4	PB7	I/O		
D5	PB4(NJTRST)	I/O	SPI1_MISO	
D6	VSS	Power		
D8	DVDD	Power		
D9	VDD50USB	Power		
D14	PA9	I/O	USART1_TX	
E2	PE4 *	I/O	GPIO_Input	R_TOE
E3	VSS	Power		
E4	PC13	I/O		
E5	PB6	I/O	FDCAN2_TX	
E6	VCAP4	Power		
E8	VSSUSB	Power		
E9	VSS	Power		
E13	PA10	I/O	USART1_RX	
F4	PE5 *	I/O	GPIO_Input	R_HEEL
F5	PE3 *	I/O	GPIO_Input	L_HEEL
F6	VSS	Power		
F7	VDD	Power		
F8	VDD33USB	Power		
F9	VDD	Power		

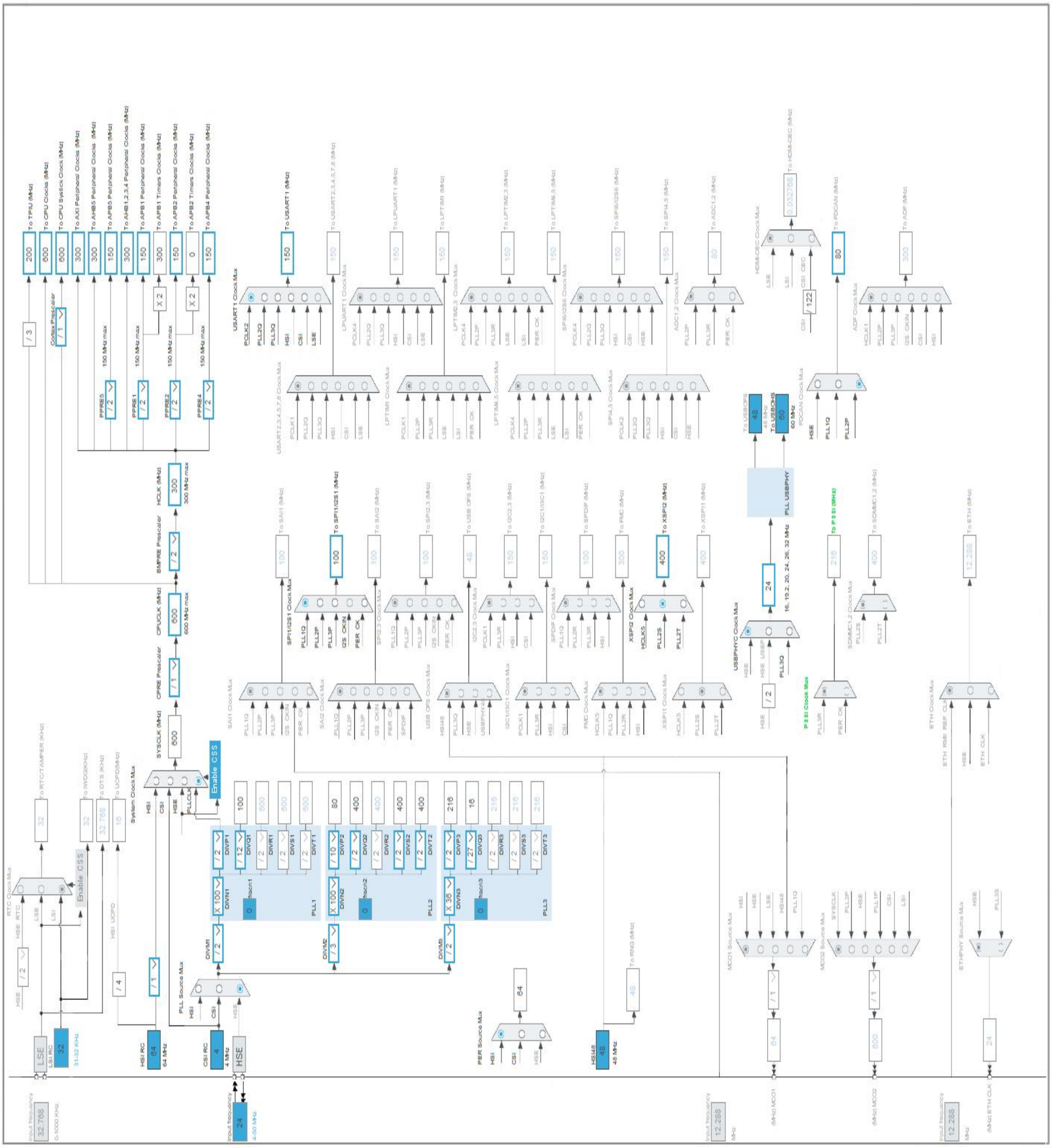
Pin Number TFBGA225 OCTO SMPS	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
F10	VDD	Power		
F11	VCAP3	Power		
F14	PN1	I/O	XSPIM_P2_NCS1	
F15	VSS	Power		
G1	VLXSMPS	Power		
G2	VFBSMPS	Power		
G6	VDD	Power		
G7	VDDLDO	Power		
G8	VSS	Power		
G9	VDDLDO	Power		
G10	VSS	Power		
G12	VSS	Power		
G13	PN3	I/O	XSPIM_P2_IO1	
G14	PN0	I/O	XSPIM_P2_DQS0	
G15	PN11	I/O	XSPIM_P2_IO7	
H1	VDDSMPS	Power		
H2	VSSSMPS	Power		
H6	VDD	Power		
H7	VSS	Power		
H8	VSS	Power		
H9	VSS	Power		
H10	VDD	Power		
H11	VSS	Power		
H12	VDDXSPI2	Power		
H13	PN10	I/O	XSPIM_P2_IO6	
H14	PN9	I/O	XSPIM_P2_IO5	
H15	PN2	I/O	XSPIM_P2_IO0	
J1	VSS	Power		
J4	NRST	Reset		
J5	VSSA	Power		
J6	VDDA	Power		
J7	VDDLDO	Power		
J8	VSS	Power		
J9	VDDLDO	Power		
J10	VSS	Power		
J11	VDDXSPI2	Power		
J12	VSS	Power		
J14	PN6	I/O	XSPIM_P2_CLK	
J15	VSS	Power		

Pin Number TFBGA225 OCTO SMPS	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
K1	PH0-OSC_IN(PH0)	I/O	RCC_OSC_IN	
K2	PH1-OSC_OUT(PH1)	I/O	RCC_OSC_OUT	
K4	VREFM	Power		
K5	VREFP	Power		
K6	VDD	Power		
K7	VSS	Power		
K8	VDD	Power		
K9	VSS	Power		
K10	VDD	Power		
K11	VCAP2	Power		
K12	VDDXSPI2	Power		
K13	PN8	I/O	XSPIM_P2_IO4	
K14	PN4	I/O	XSPIM_P2_IO2	
K15	PN5	I/O	XSPIM_P2_IO3	
L6	VSS	Power		
L7	VCAP1	Power		
L8	VDDXSPI1	Power		
L9	VDD	Power		
L10	VDDXSPI1	Power		
L11	VSS	Power		
M8	VSS	Power		
M9	VSS	Power		
M10	VDDXSPI1	Power		
M11	PD8	I/O		
M15	VSS	Power		
P9	VSS	Power		
P13	PD9	I/O		
P14	PD13	I/O		
R1	VSS	Power		
R6	VSS	Power		
R12	VSS	Power		
R14	PD10	I/O		
R15	VSS	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration

RIF configuration not available for STM32H7S3L8Hx



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32H7
Line	STM32H7R3/7S3
MCU	STM32H7S3L8Hx
Datasheet	DS000000_Rev1

1.2. Parameter Selection

Temperature	25
Vdd	3.0

1.3. Battery Selection

Battery	Alkaline(9V)
Capacity	625.0 mAh
Self Discharge	0.3 %/month
Nominal Voltage	9.0 V
Max Cont Current	200.0 mA
Max Pulse Current	0.0 mA
Cells in series	1
Cells in parallel	1

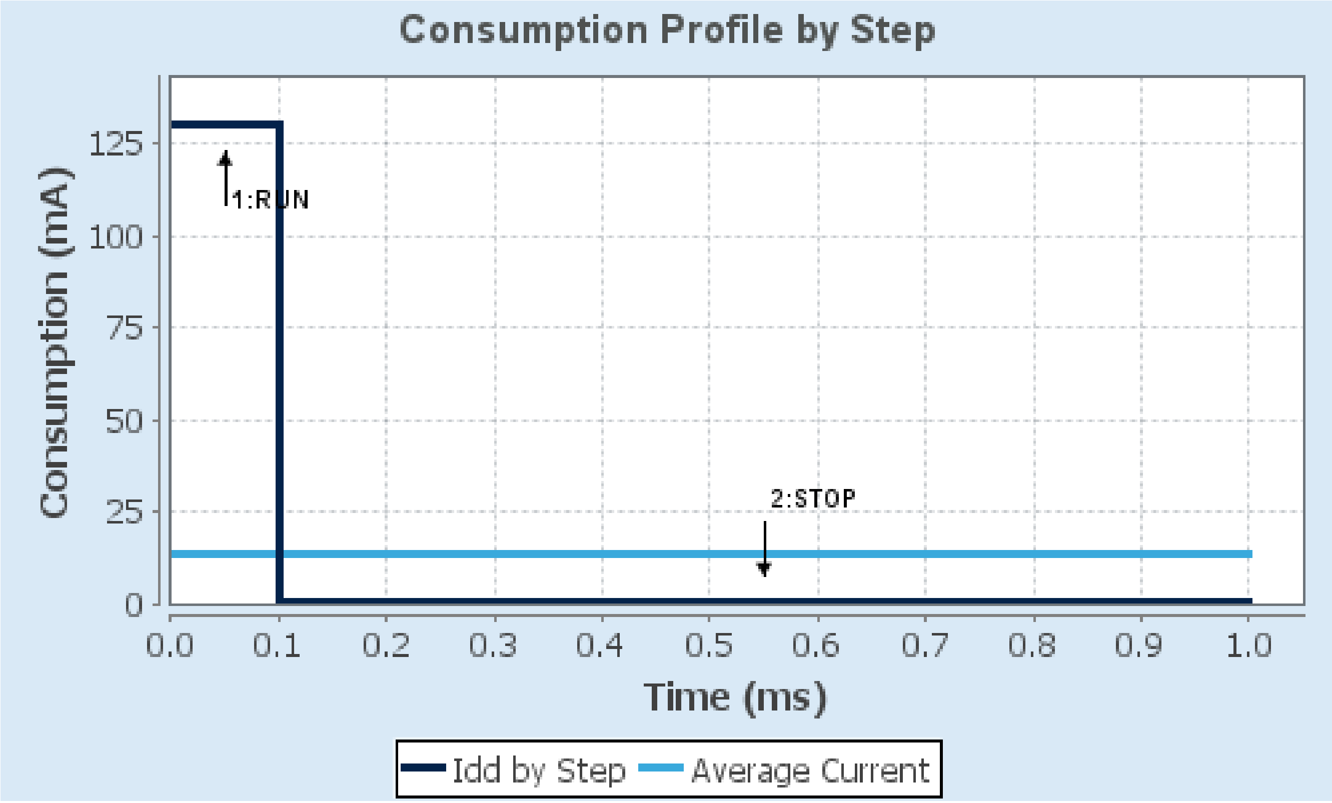
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	3.0	3.0
Voltage Source	Battery	Battery
Range	VOS-HIGH	SVOS-LOW/SMPS
Fetch Type	AXISRAM3/FlashMode-ON/Cache/AlgoType-CoreMark	FLASH
CPU Frequency	600 MHz	0 Hz
Clock Configuration	HSE BYP PLL	ALL_CLOCKS_OFF
Clock Source Frequency	8 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	130 mA	265 µA
Duration	0.1 ms	0.9 ms
DMIPS	1284.0	0.0
Ta Max	107.45	124.96
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	13.24 mA
Battery Life	1 day, 23 hours	Average DMIPS	128.40001 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
125
100
75
50
25
0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	nexus_first
Project Folder	C:\Users\visma\Downloads\nexus_first
Toolchain / IDE	CMake
Firmware Package Name and Version	STM32Cube FW_H7RS V1.3.0
Application Structure	Advanced
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls Boot

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_GPDMA1_Init	GPDMA1
4	MX_SBS_Init	SBS
5	MX_XSPI2_Init	XSPI2
6	MX_EXTMEM_MANAGER_Init	EXTMEM_MANAGER

2.4. Advanced Settings - Generated Function Calls Appli

Rank	Function Name	Peripheral Instance Name

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_GPDMA1_Init	GPDMA1
4	MX_FDCAN1_Init	FDCAN1
5	MX_FDCAN2_Init	FDCAN2
6	MX_SPI1_Init	SPI1
7	MX_USART1_UART_Init	USART1
8	MX_TIM2_Init	TIM2
9	MX_TIM6_Init	TIM6
10	MX_USB_DEVICE_Init	USB_DEVICE
11	MX_XSPI2_Init	XSPI2

2.5. Advanced Settings - Generated Function Calls ExtMemLoader

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_GPDMA1_Init	GPDMA1
4	MX_SBS_Init	SBS
5	MX_XSPI2_Init	XSPI2
6	MX_EXTMEM_MANAGER_Init	EXTMEM_MANAGER

3. Peripherals and Middlewares Configuration

3.1. CORTEX_M7_BOOT

3.1.1. Parameter Settings:

Speculation default mode Settings:

Speculation default mode	**Enabled ***

Cortex Interface Settings:

CPU ICache	**Enabled ***
CPU DCache	**Enabled ***

Cortex Memory Protection Unit Control Settings:

MPU Control Mode	Background Region Privileged accesses only + MPU Disabled during hard fault, NMI and FAULTMASK handlers

Cortex Memory Protection Unit Region 0 Settings:

MPU Region	Enabled
MPU Region Base Address	**0x0 ***
MPU Region Size	4GB
MPU SubRegion Disable	**0x87 ***
MPU TEX field level	level 0
MPU Access Permission	ALL ACCESS NOT PERMITTED
MPU Instruction Access	DISABLE
MPU Shareability Permission	**DISABLE ***
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 1 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x70000000 ***
MPU Region Size	**128MB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	**level 1 ***
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	ENABLE
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	**ENABLE ***
MPU Bufferable Permission	**ENABLE ***

Cortex Memory Protection Unit Region 2 Settings:

MPU Region	**Enabled ***
MPU Region Base Address	**0x0 ***

MPU Region Size	**16KB ***
MPU SubRegion Disable	**0x0 ***
MPU TEX field level	**level 1 ***
MPU Access Permission	**ALL ACCESS PERMITTED ***
MPU Instruction Access	**DISABLE ***
MPU Shareability Permission	DISABLE
MPU Cacheable Permission	DISABLE
MPU Bufferable Permission	DISABLE

Cortex Memory Protection Unit Region 3 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 4 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 5 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 6 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 7 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 8 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 9 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 10 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 11 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 12 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 13 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 14 Settings:

MPU Region	Disabled

Cortex Memory Protection Unit Region 15 Settings:

MPU Region	Disabled

3.2. FDCAN1

mode: Activated

3.2.1. Parameter Settings:

Basic Parameters:

Clock Divider	Divide kernel clock by 1
Frame Format	**FD mode with BitRate Switching ***
Mode	Normal mode
Auto Retransmission	**Enable ***
Transmit Pause	Disable
Protocol Exception	**Enable ***
Nominal Sync Jump Width	**10 ***
Data Prescaler	1
Data Sync Jump Width	**4 ***
Data Time Seg1	**11 ***
Data Time Seg2	**4 ***
Std Filters Nbr	**1 ***
Ext Filters Nbr	0
Tx Fifo Queue Mode	FIFO mode

Bit Timings Parameters:

Nominal Prescaler	**1 ***
Nominal Time Quantum	**12.5 ***
Nominal Time Seg1	**63 ***
Nominal Time Seg2	**16 ***
Nominal Time for one Bit	1000
Nominal Baud Rate	**1000000 ***

3.3. FDCAN2

mode: Activated

3.3.1. Parameter Settings:

Basic Parameters:

Clock Divider	Divide kernel clock by 1
Frame Format	**FD mode with BitRate Switching ***
Mode	Normal mode
Auto Retransmission	**Enable ***
Transmit Pause	Disable
Protocol Exception	

	Enable *
Nominal Sync Jump Width	**10 ***
Data Prescaler	1
Data Sync Jump Width	**4 ***
Data Time Seg1	**11 ***
Data Time Seg2	**4 ***
Std Filters Nbr	**1 ***
Ext Filters Nbr	0
Tx Fifo Queue Mode	FIFO mode
Bit Timings Parameters:	
Nominal Prescaler	**1 ***
Nominal Time Quantum	**12.5 ***
Nominal Time Seg1	**63 ***
Nominal Time Seg2	**16 ***
Nominal Time for one Bit	1000
Nominal Baud Rate	**1000000 ***

3.4. NUCLEO-H7S3L8
mode: Human Machine Interface
3.4.1. Human Machine Interface:

Led:	
USER LED GREEN (LD1)	**true ***
USER LED YELLOW (LD2)	**true ***
USER LED RED (LD3)	**true ***
Button:	
USER BUTTON	**Mode EXTI ***
VCOM:	
Virtual Com Port	**true ***
Demonstration code:	
Generate demonstration code	Disabled

3.5. RCC
High Speed Clock (HSE): Crystal/Ceramic Resonator

3.5.1. Parameter Settings:

Power Parameters:	
SupplySource	**PWR_LDO_SUPPLY ***
Power Regulator Voltage Scale	Power Regulator Voltage Scale 0
RCC Parameters:	
TIM Prescaler Selection	Disabled
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
HSI Calibration Value	64
System Parameters:	
VDD voltage (V)	3.3
Flash Latency(WS)	7 WS (8 CPU cycle)
PLL range Parameters:	
PLL1 input frequency range	Between 8 and 16 MHz
PLL2 input frequency range	Between 8 and 16 MHz
PLL1 clock Output range	High VCO range
PLL2 clock Output range	High VCO range

3.6. SBS
mode: Activated
3.6.1. Parameter Settings:

Basic parameters:	
IO HSLV FOR XSPIM1	DISABLE
IO HSLV FOR XSPIM2	**ENABLE ***
IO HSLV FOR OTHER IO	DISABLE

3.7. SPI1
Mode: Full-Duplex Master
3.7.1. Parameter Settings:

Basic Parameters:	
Frame Format	Motorola
Data Size	**16 Bits ***
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	**16 ***
Baud Rate	**6.25 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	**2 Edge ***

CRC Parameters:

CRC Calculation	Disabled

Advanced Parameters:

NSSP Mode	**Disabled ***
NSS Signal Type	Software
Fifo Threshold	Fifo Threshold 01 Data
Nss Polarity	Nss Polarity Low
Master Ss Idleness	00 Cycle
Master Inter Data Idleness	00 Cycle
Master Receiver Auto Susp	Disable
Master Keep Io State	Master Keep Io State Disable
IO Swap	Disabled
Ready Master Management	Internal
Ready Signal Polarity	High

3.8. SYS

Timebase Source: SysTick

3.8.1. Core(s) Settings:

Context(s):	Boot

3.9. TIM2

Clock Source : Internal Clock

3.9.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	**299 ***
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 32 bits value)	4294967295
Internal Clock Division (CKD)	No Division

auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Master/Slave Mode (MSM bit)	Disable (Trigger input effect not delayed)
Trigger Event Selection TRGO	Reset (UG bit from TIMx_EGR)

3.10. TIM6
mode: Activated
3.10.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	**299 ***
Counter Mode	Up
Dithering	Disable
Counter Period (AutoReload Register - 16 bits value)	**999 ***
auto-reload preload	Disable

Trigger Output (TRGO) Parameters:

Trigger Event Selection	Reset (UG bit from TIMx_EGR)

3.11. USART1
Mode: Asynchronous
3.11.1. Parameter Settings:

Basic Parameters:

Baud Rate	**3000000 ***
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	**8 Samples ***
Single Sample	Disable
ClockPrescaler	1
Fifo Mode	Disable
Txfifo Threshold	1 eighth full configuration
Rxfifo Threshold	1 eighth full configuration

Advanced Features:

Auto Baudrate	Disable

TX Pin Active Level Inversion	Disable
RX Pin Active Level Inversion	Disable
Data Inversion	Disable
TX and RX Pins Swapping	Disable
Overrun	Enable
DMA on RX Error	Enable
MSB First	Disable

3.12. USB_OTG_HS
Internal HS Phy: Device_Only
3.12.1. Parameter Settings:

Speed	Device High Speed 480MBit/s
Enable internal IP DMA	**Enabled ***
Physical interface	Internal HS Phy
Low power	Disabled
Link Power Management	Disabled
Use dedicated end point 1 interrupt	Disabled
VBUS sensing	Disabled
Signal start of frame	Disabled
OTG PHY reference clock selection:	
Ref Clock Selection	**24 Mhz ***

3.13. XSPI2
Mode: Octo SPI
Port: Port2 Octo
Chip Select Override: NCS1 -- Port2 --
3.13.1. Parameter Settings:

Generic:	
Fifo Threshold	**4 ***
Memory Mode	Disable
Memory Type	**Macronix ***
Memory Size	**256 MBits ***
Chip Select High Time Cycle	**2 ***
Free Running Clock	Disable
Clock Mode	Low

Wrap Size	Not Supported
Clock Prescaler	**3 ***
Sample Shifting	None
Chip Select Boundary	Disabled
Maximum Transfer	0
Refresh Rate	0
Memory Select	NCS1

3.14. EXTMEM_MANAGER
mode: Activate External Memory Manager
3.14.1. Boot usecase:

Boot:	
Select boot code generation	**true ***
Selection of the boot system	Execute In Place
XIP:	
select the memory	Memory 1

3.14.2. Memory 1:

Select driver:	
Select the memory driver	EXTMEM_NOR_SFDP
Configuration:	
Memory Instance	XSPI2
Number of memory data lines	**EXTMEM_LINK_CONFIG_8LINES ***

3.14.3. Memory 2:

Select driver:	
Select the memory driver	NONE

3.15. USB_DEVICE
Class For HS IP: Communication Device Class (Virtual Port Com)
3.15.1. Parameter Settings:

Basic Parameters:

USBD_MAX_NUM_INTERFACES (Maximum number of supported interfaces)	1
USBD_MAX_NUM_CONFIGURATION (Maximum number of supported configuration)	1
USBD_MAX_STR_DESC_SIZ (Maximum size for the string descriptors)	512
USBD_SELF_POWERED (Enabled self power)	Enabled
USBD_DEBUG_LEVEL (USBD Debug Level)	0: No debug message
USBD_LPM_ENABLED (Link Power Management)	**0: Link Power Management not supported ***

Class Parameters:

USB CDC Rx Buffer Size	2048
USB CDC Tx Buffer Size	2048

3.15.2. Device Descriptor:

Device Descriptor:

VID (Vendor IDentifier)	1155
LANGID_STRING (Language Identifier)	English(United States)
MANUFACTURER_STRING (Manufacturer Identifier)	STMicroelectronics

Device Descriptor HS:

PID (Product IDentifier)	22336
PRODUCT_STRING (Product Identifier)	STM32 Virtual ComPort
CONFIGURATION_STRING (Configuration Identifier)	CDC Config
INTERFACE_STRING (Interface Identifier)	CDC Interface

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
FDCAN1	PB8	FDCAN1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PD1	FDCAN1_TX	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
FDCAN2	PB5	FDCAN2_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
	PB6	FDCAN2_TX	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
RCC	PH0-OSC_IN(PH0)	RCC_OSC_IN	n/a	n/a	n/a	
	PH1-OSC_OUT(PH1)	RCC_OSC_OUT	n/a	n/a	n/a	
SPI1	PD7	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PB3(JTDO/TRACESWO)	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PB4(NJTRST)	SPI1_MISO	Alternate Function Push Pull	**Pull-up ***	Low	
USART1	PA9	USART1_TX	Alternate Function Push Pull	No pull-up and no pull-down	**High ***	
	PA10	USART1_RX	Alternate Function Push Pull	No pull-up and no pull-down	Low	
USB_OTG_HS	PM6	USB_OTG_HS_DP	n/a	n/a	n/a	
	PM5	USB_OTG_HS_DM	n/a	n/a	n/a	
XSPI2	PN1	XSPIM_P2_NCS1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN3	XSPIM_P2_IO1	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN0	XSPIM_P2_DQS0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN11	XSPIM_P2_IO7	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN10	XSPIM_P2_IO6	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN9	XSPIM_P2_IO5	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN2	XSPIM_P2_IO0	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN6	XSPIM_P2_CLK	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN8	XSPIM_P2_IO4	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN4	XSPIM_P2_IO2	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
	PN5	XSPIM_P2_IO3	Alternate Function Push Pull	No pull-up and no pull-down	Very High	
GPIO	PE2	GPIO_Input	Input mode	**Pull-up ***	n/a	L_TOE
	PF1	GPIO_Output	Output Push Pull	**Pull-up ***	Low	enc_cs
	PE4	GPIO_Input	Input mode	**Pull-up ***	n/a	R_TOE

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PE5	GPIO_Input	Input mode	**Pull-up ***	n/a	R_HEEL
	PE3	GPIO_Input	Input mode	**Pull-up ***	n/a	L_HEEL

4.2. GPDMA1

Channel 3 - 2 Words Internal FIFO : Standard Request Mode
Channel 2 - 2 Words Internal FIFO : Standard Request Mode
Channel 1 - 2 Words Internal FIFO : Standard Request Mode
Channel 0 - 2 Words Internal FIFO : Standard Request Mode

4.2.1. All Channels:

Channel 0:

Request GPDMA1_REQUEST_SPI1_RX

Channel 1:

Request GPDMA1_REQUEST_SPI1_TX

Channel 2:

Request GPDMA1_REQUEST_USART1_RX

Channel 3:

Request GPDMA1_REQUEST_USART1_TX

4.2.2. SECURITY:

CH3:

Enable Channel as Privileged NON PRIVILEDGED

CH2:

Enable Channel as Privileged NON PRIVILEDGED

CH1:

Enable Channel as Privileged NON PRIVILEDGED

CH0:

Enable Channel as Privileged NON PRIVILEDGED

4.2.3. CH3:

Circular configuration:

Circular Mode Disable

Request Configuration:

Request	**USART1_TX ***
DMA Handle in IP Structure	hdmatx
Block HW request protocol	Single/Burst Level
Channel configuration:	
Priority	Low
Transaction Mode	Normal
Direction	**Memory To Peripheral ***
Source Data Setting:	
Source Address Increment After Transfer	**Enabled ***
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0
Destination Data Setting:	
Destination Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0
Data Handling:	
Data Handling Configuration	Disable
Trigger:	
Trigger Configuration	Disable
Transfer Event Configuration:	
Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block

4.2.4. CH2:

Circular configuration:	
Circular Mode	**Enable ***
Circular Port	Port 0
Request Configuration:	
Request	**USART1_RX ***
DMA Handle in IP Structure	hdmarx
Block HW request protocol	Single/Burst Level
Channel configuration:	
Priority	**High ***
Transaction Mode	Normal
Direction	Peripheral To Memory
Node Type	GPDMA Linear Addressing
Source Data Setting:	

Source Address Increment After Transfer	Disabled
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	**Enabled ***
Data Width	Byte
Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable

Trigger:

Trigger Configuration	Disable

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block

Channel Configuration for Linked List:

Execution Mode (circular/linear) of the Linked List	Circular
Linked List Execution Mode	The List is fully executed

4.2.5. CH1:

Circular configuration:

Circular Mode	Disable

Request Configuration:

Request	**SPI1_TX/I2S1_TX ***
DMA Handle in IP Structure	hdmatx
Block HW request protocol	Single/Burst Level

Channel configuration:

Priority	**Medium ***
Transaction Mode	Normal
Direction	**Memory To Peripheral ***

Source Data Setting:

Source Address Increment After Transfer	**Enabled ***
Data Width	**Half Word ***
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	Disabled
Data Width	**Half Word ***

Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable

Trigger:

Trigger Configuration	Disable

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block

4.2.6. CH0:

Circular configuration:

Circular Mode	Disable

Request Configuration:

Request	**SPI1_RX/I2S1_RX ***
DMA Handle in IP Structure	hdmarx
Block HW request protocol	Single/Burst Level

Channel configuration:

Priority	**High ***
Direction	Peripheral To Memory
Transaction Mode	Normal

Source Data Setting:

Source Address Increment After Transfer	Disabled
Data Width	**Half Word ***
Burst Length	1
Allocated Port for Transfer	Port 0

Destination Data Setting:

Destination Address Increment After Transfer	**Enabled ***
Data Width	**Half Word ***
Burst Length	1
Allocated Port for Transfer	Port 0

Data Handling:

Data Handling Configuration	Disable

Trigger:

Trigger Configuration	Disable

Transfer Event Configuration:

Transfer Event Generation	The TC (and the HT) event is generated at the (respectively half) end of each block

4.3. HPDMA1

4.3.1. Core(s) Settings:

Context(s):	Boot Application ExternalMemoryLoader
Initialized Context:	Application
Power Domain:	

4.4. LINKEDLIST

4.4.1. Core(s) Settings:

Context(s):	Boot Application ExternalMemoryLoader
Initialized Context:	Application
Power Domain:	

4.5. NVIC configuration

4.5.1. NVIC1

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
EXTI line13 interrupt	true	0	0
GPDMA1 Channel 0 global interrupt	true	2	0
GPDMA1 Channel 1 global interrupt	true	2	0
GPDMA1 Channel 2 global interrupt	true	4	0
GPDMA1 Channel 3 global interrupt	true	4	0
TIM6 global interrupt	true	1	0
SPI1 global interrupt	true	2	0
USART1 global interrupt	true	4	0
USB OTG HS interrupt	true	6	0
FDCAN1 interrupt 0	true	3	0
FDCAN2 interrupt0	true	3	0
PVD and PVM interrupts through EXTI line	unused		
RCC global interrupt	unused		
FLASH interrupts	unused		
FPU global interrupt	unused		
TIM2 global interrupt	unused		
XSPI2 global interrupt	unused		
FDCAN1 interrupt 1	unused		
FDCAN2 interrupt 1	unused		

4.5.2. NVIC1 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
EXTI line13 interrupt	false	true	true
GPDMA1 Channel 0 global interrupt	false	true	true
GPDMA1 Channel 1 global interrupt	false	true	true
GPDMA1 Channel 2 global interrupt	false	true	true
GPDMA1 Channel 3 global interrupt	false	true	true
TIM6 global interrupt	false	true	true
SPI1 global interrupt	false	true	true
USART1 global interrupt	false	true	true
USB OTG HS interrupt	false	true	true
FDCAN1 interrupt 0	false	true	true
FDCAN2 interrupt0	false	true	true

4.5.3. NVIC2

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
EXTI line13 interrupt	true	0	0
PVD and PVM interrupts through EXTI line	unused		
RCC global interrupt	unused		
FLASH interrupts	unused		
FPU global interrupt	unused		
XSPI2 global interrupt	unused		

4.5.4. NVIC2 Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true
EXTI line13 interrupt	false	true	true

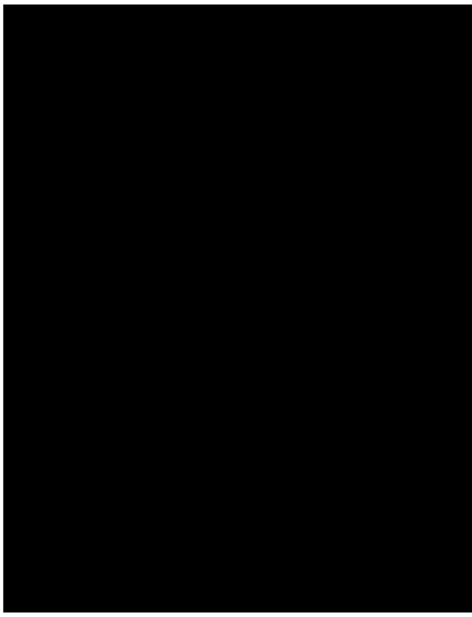
*** User modified value**

5. System Views

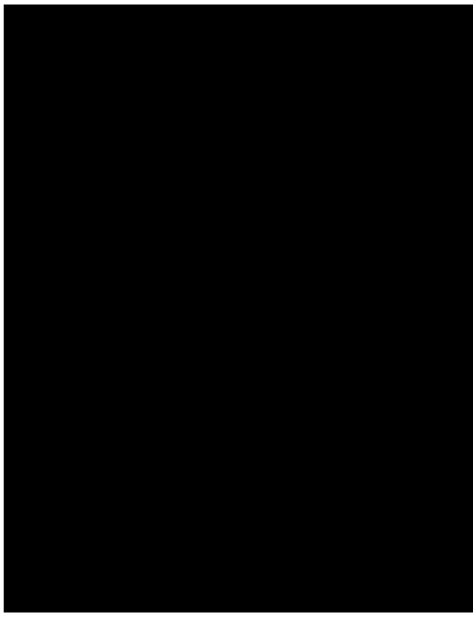
5.1. Category view

5.1.1. Current

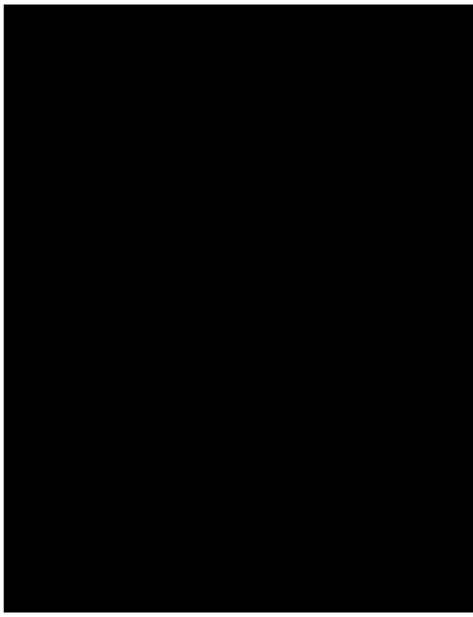
5.1.2. Without filters



5.2. Context Execution view



5.3. Context Initialization view



6. Docs & Resources

Type	Link